

Data Quality Assessment & Intake Specification Proposal

UCI Online Retail II Dataset — 1,067,371 Transactions | 2009–2011

Prepared by **Ahmed Maala** | Application: Data Integration Intern | PRYZM Solutions GmbH, Germany



1. Data Quality Assessment

The UCI Online Retail II dataset contains **1,067,371 transaction records** spanning two years of wholesale activity (December 2009 – December 2011). This assessment mirrors the type of raw data PRYZM Solutions receives from industrial and pharmaceutical clients. Six distinct quality dimensions were evaluated, revealing issues that range from critical data gaps to minor formatting inconsistencies.

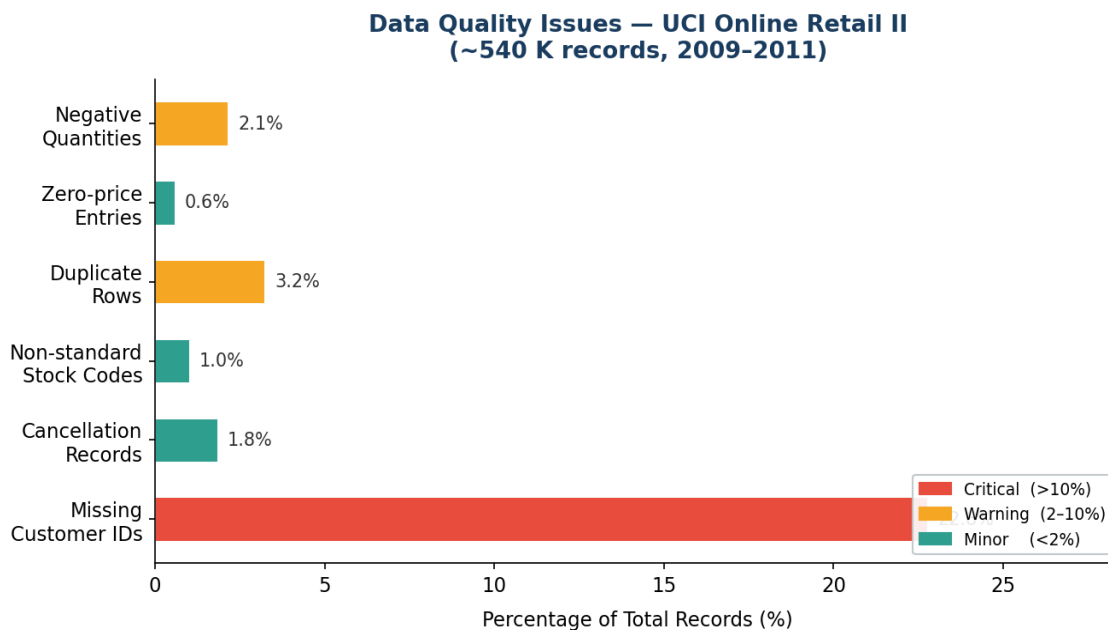


Figure 1 — Data quality issues ranked by severity. Red = Critical (>10%), Amber = Warning (2–10%), Teal = Minor (<2%).

1.1 Completeness — Missing Values

The most critical finding is that **22.77% of all records lack a Customer ID** (243,007 rows). This completely prevents any customer-level analysis, including cohort tracking, lifetime value calculation, or churn prediction. A secondary gap exists in the Description field (0.41%, 4,382 rows), which affects product categorisation pipelines.

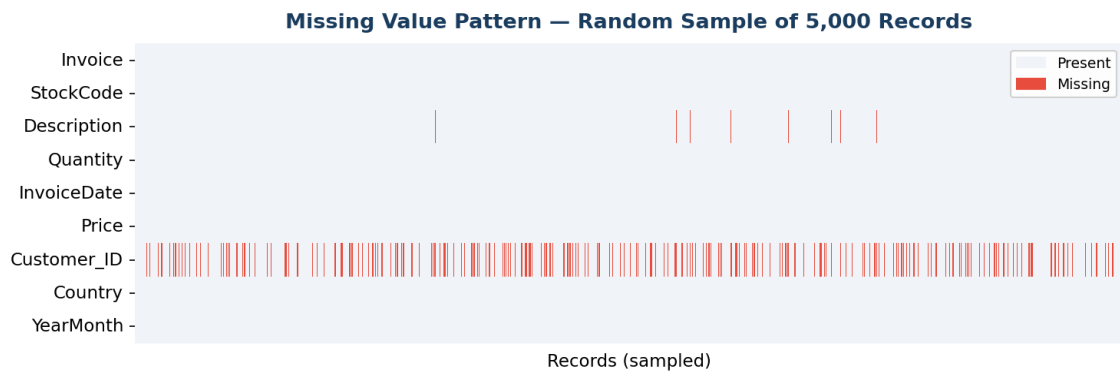


Figure 2 — Missing value pattern across all columns in a random sample of 5,000 records. The dense red band on Customer_ID confirms the systemic nature of this gap.

1.2 Accuracy — Anomalous Values

Issue	Count	% of Total	Severity
Negative Quantities	22,950	2.15%	Warning
Zero-Price Entries	6,202	0.58%	Minor
Non-standard Stock Codes	10,739	1.01%	Minor
Negative Prices	5	<0.01%	Minor

Negative quantities (22,950 records) are the most impactful accuracy issue. While some correspond to legitimate return transactions, others appear to be data entry errors unrelated to any cancellation invoice. Non-standard stock codes such as "POST", "DOT", and "BANK CHARGES" represent operational charges rather than physical products and must be excluded from inventory analytics.

1.3 Consistency — Duplicates & Cancellations

Exact duplicate rows account for **3.22% (34,335 records)**. These are likely caused by repeated API calls or batch re-submissions and will inflate revenue figures if not removed at the staging layer. Cancellation records (1.83%, 19,494 rows, identified by an invoice prefix of "C") are currently mixed with standard sales, making it impossible to compute accurate net revenue without additional filtering.

1.4 Temporal & Geographic Distribution

Transaction volume shows a clear **seasonal peak in Q4** each year, consistent with retail demand patterns. The dataset is heavily skewed toward the United Kingdom (92% of all records), with Germany, France, and Ireland representing the next largest markets — directly relevant to PRYZM's European client base.

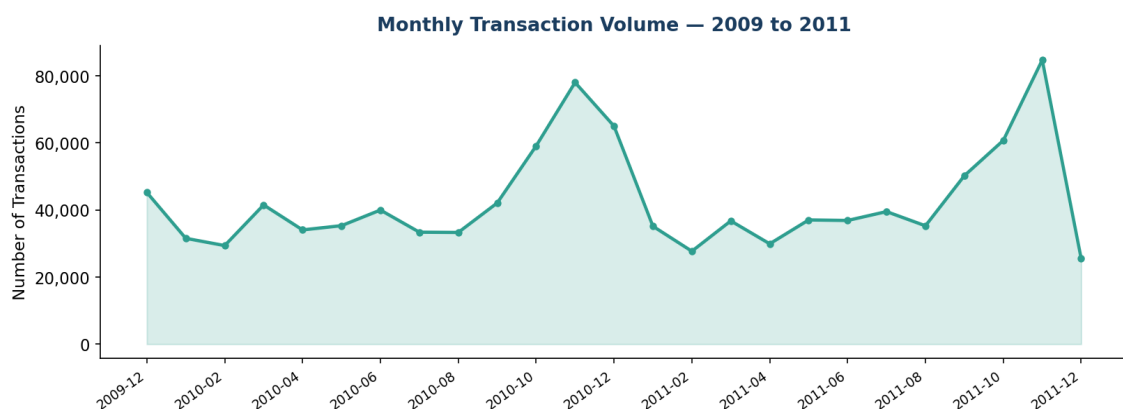


Figure 3 — Monthly transaction volume. Q4 spikes are consistent across both years.

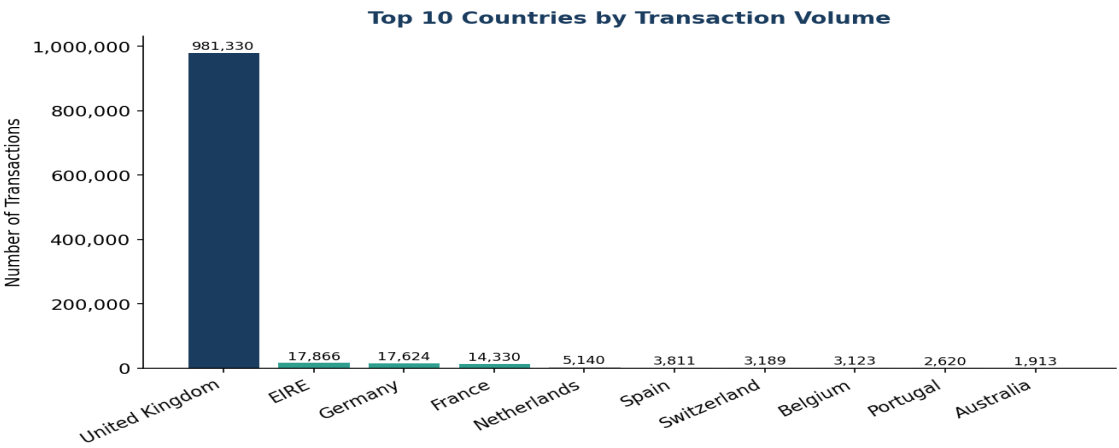


Figure 4 — Top 10 countries by transaction volume. UK dominates at 92%; Germany ranks third.

2. Proposed Intake Specification

To prevent the identified issues at the source and ensure clean, analysis-ready data from day one, I propose the following intake specification for client data submissions. This specification is designed to be lightweight enough for clients to implement quickly, while providing PRYZM's pipeline with reliable, validated inputs.

2.1 Required Fields & Accepted Formats

Field Name	Type	Requirement	Validation Rule
InvoiceNo	String (20)	Mandatory	No spaces; prefix "C" reserved for returns
StockCode	String (15)	Mandatory	Pattern: ^\d{5}[A-Z]?\$
Description	String (255)	Optional	Free text; UTF-8 encoded
Quantity	Integer	Mandatory	Must be ≠ 0; negative = return
InvoiceDate	Datetime	Mandatory	ISO 8601: YYYY-MM-DD HH:MM:SS
UnitPrice	Decimal(10,2)	Mandatory	Must be ≥ 0
CustomerID	String (20)	Mandatory	Non-null; unique per customer
Country	String	Optional	ISO 3166-1 alpha-2 preferred

2.2 Automated Pre-Ingestion Validation Checks

The following checks should be executed automatically at the staging layer before any record reaches the data warehouse. Each check produces a structured quality report that is logged and, where thresholds are exceeded, triggers an alert to the client.

#	Check Name	Logic	Action on Failure
1	Null Rate Guard	CustomerID null rate > 5%	Reject batch; notify client
2	Type Enforcement	Quantity not integer OR UnitPrice not numeric	Reject affected rows
3	Price Sanity	UnitPrice < 0	Flag & quarantine row
4	Duplicate Removal	Exact row-level duplicate	Drop silently; log count
5	Return Separation	Quantity < 0 OR Invoice starts with "C"	Route to returns_fact table
6	StockCode Pattern	Does not match ^\d{5}[A-Z]?\$	Flag as operational code

3. Top Priority Improvement

Priority #1: Implement a Returns & Cancellations Separation Protocol

The Problem. Returns (negative quantities) and cancellations (Invoice prefix "C") are currently co-mingled with standard sales records. This single issue cascades into at least four downstream failures: gross revenue is overstated, inventory levels are incorrect, customer purchase frequency is inflated, and any machine-learning model trained on this data will learn from corrupted signals.

The Solution. Introduce a two-table architecture at the staging layer: a **sales_fact** table containing only clean, positive-quantity transactions, and a **returns_fact** table containing all negative-quantity and cancellation records. A reconciliation step then links each return to its original sale using a SQL Window Function or Pandas merge on (CustomerID, StockCode, |Quantity|), enabling accurate net revenue and return-rate reporting.

Implementation Sketch (Python/Pandas):

```
df_sales = df[df["Quantity"] > 0 & ~df["Invoice"].str.startswith("C")] df_returns =  
df[df["Quantity"] < 0 | df["Invoice"].str.startswith("C")] # Reconcile returns to original  
invoices df_net = df_sales.merge( df_returns.rename(columns={"Quantity": "ReturnQty"}),  
on=["CustomerID", "StockCode"], how="left" ).assign(NetQty=lambda x: x["Quantity"] +  
x["ReturnQty"].fillna(0))
```

Why this matters most. Resolving this single issue immediately delivers clean data to every downstream consumer — dashboards, forecasting models, and financial reports — without requiring any schema changes or client-side modifications. It is the highest-leverage action available and can be deployed within one sprint.

Ahmed Maala | contact@ahmedmaala.com | LinkedIn: linkedin.com/in/ahmedmaala | GitHub: github.com/ahmedmaala